

Homework #1

Due Friday January 17, before midnight (CT)

Submit to Canvas

Read Chapters 2 of Part I of Strogatz.

The following problems are from Strogatz Second Edition, the one with the black cover. Scans of the exercises in chapters 2 can be found on Canvas in the files section, in a folder called Problem Sets.

Solutions to the following 8 problems should be written up carefully and turned in online. The problems marked by a * are worth 10 points, and the others are worth 5 points. To get full credit on a problem, your solution needs to be not only correct, but also clearly presented.

2.2.7, 2.2.13*, 2.3.1, 2.3.4*, 2.4.9, 2.5.2, 2.5.4, 2.8.6 (a)-(b)

For 2.2.7: Comment not only on the existence of the fixed point(s) but also on their stability.

For 2.3.1(b): For this part you first derive a differential equation that x satisfies, and then you solve that differential equation for $x(t)$. You should be able to compare with your answer to part (a), by finally converting the solution from the x variable back to the N variable.

Extra Credit: complete 2.8.6. (Note that I only assigned the first two parts of this problem above. I omitted 2.8.6 (c-d).)